$$f(\theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos(n\theta) + b_n \sin(n\theta)\right] \qquad F\text{-}S$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\cos(n\theta)\,d\theta$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\sin(n\theta)\,d\theta$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(\theta)\,d\theta$$

1) F-S is periodic on $-\pi < \theta < \pi$

2) what fn's have F-S? all integrals are finite

$$|a_n \cos(n\theta) + b_n \sin(n\theta)| \le |a_n| + |b_n|$$

3) when does F-S converge to func?
if $f(\theta)$ piecewise smooth on $-\pi \le \theta \le \pi$

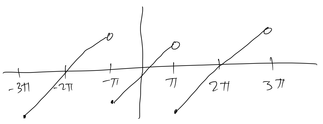
piecewise smooth: The interval can be broken up into pieces wher $f \in C^1$

ok.

not ok

when $f(\theta)$ is piecewise smooth
F-S converges to $f(\theta)$ where f is cont.
and $\frac{1}{2}\left(f(\theta^+) + f(\theta^-)\right)$ o.w.

<u>periodic extension</u>

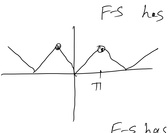


$f : [0, \pi) \to \mathbb{R}$

even: $f(-x) = f(x)$ F-S has only cos terms
ex: $\cos(x)$, $|x|$

odd: $f(-x) = -f(x)$ F-S has only sin terms
ex: $\sin(x)$

even extension: F-S has only cos terms



odd extension F-S has only sin terms

F-S: $$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)\right]$$

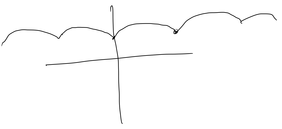
$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx$$

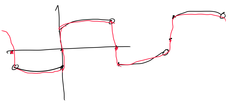
$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx$$

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\,dx$$

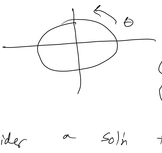
Piecewise smooth give uniform convergence,
which allows $\sum_{n=1}^{\infty}\frac{d}{dx} = \frac{d}{dx}\sum_{n=1}^{\infty}$

Gibbs phenomena from truncation of F-S





Sec 7.7. Dirichlet problem on unit disc



$-\pi \le \theta \le \pi$

$$\begin{cases} \Delta u = 0 & 0 \le r < 1,\ -\pi \le \theta \le \pi \\ u(1,\theta) = f(\theta) \end{cases}$$

consider a soln that is a superposition of $r^n \cos(n\theta)$ and $r^n \sin(n\theta)$

Claim: $$u(r,\theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n r^n \cos(n\theta) + b_n r^n \sin(n\theta)\right)$$

$0 \le r < 1$ is harmonic

$$\Delta u(r,\theta) = \Delta\left(\frac{a_0}{2}\right) + \underbrace{\Delta \sum_{n=1}^{\infty}\left(a_n r^n \cos(n\theta) + b_n r^n \sin(n\theta)\right)}$$

$= 0 +$

can we interchange Δ & Σ?
yes, for $r < 1$ (Wiestrass m-test / geometric series)

$$\Rightarrow \Delta u(r,\theta) = \sum_{n=1}^{\infty}\Delta\left[a_n r^n \cos(n\theta) + b_n r^n \sin(n\theta)\right] = 0$$

$|a_n|, |b_n| \le M \quad \forall n$

$$\sum_{n=1}^{\infty}\left|\left[a_n r^n \cos(n\theta) + b_n r^n \sin(n\theta)\right]\right| \le \sum_{n=1}^{\infty}\left|\left[M r^n + M r^n\right]\right|$$